

Implantable Cardioverter Defibrillators (ICDs)

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Tracking Information

Publication Number

100-3

Manual Section Number

20.4

Manual Section Title

Implantable Cardioverter Defibrillators (ICDs)

Version Number

5

Effective Date of this Version

07/31/2023

Implementation Date

07/31/2023

Description Information

Benefit Category

Prosthetic Devices

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

An implantable cardioverter defibrillator (ICD) is an electronic device designed to diagnose and treat life-threatening ventricular tachyarrhythmias.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective for services performed on or after February 15, 2018 CMS has determined that the evidence is sufficient to conclude that the use of ICDs, (also referred to as defibrillators) is

reasonable and necessary:

1. Patients with a personal history of sustained ventricular tachyarrhythmia (VT) or cardiac arrest due to ventricular fibrillation (VF). Patients must have demonstrated:
 - An episode of sustained VT, either spontaneous or induced by an electrophysiology (EP) study, not associated with an acute myocardial infarction (MI) and not due to a transient or reversible cause; or
 - An episode of cardiac arrest due to VF, not due to a transient or reversible cause.
2. Patients with a prior MI and a measured left ventricular ejection fraction (LVEF) ≤ 0.30 .

Patients must not have:

- New York Heart Association (NYHA) classification IV heart failure; or,
- Had a coronary artery bypass graft (CABG), or percutaneous coronary intervention (PCI) with angioplasty and/or stenting, within the past 3 months; or,
- Had an MI within the past 40 days; or,
- Clinical symptoms and findings that would make them a candidate for coronary revascularization.

For these patients identified in B2, a formal shared decision making encounter must occur between the patient and a physician (as defined in Section 1861(r)(1) of the Social Security Act (the Act)) or qualified non-physician practitioner (meaning a physician assistant, nurse practitioner, or clinical nurse specialist as defined in §1861(aa)(5) of the Act) using an evidence-based decision tool on ICDs prior to initial ICD implantation. The shared decision making encounter may occur at a separate visit.

3. Patients who have severe, ischemic, dilated cardiomyopathy but no personal history of sustained VT or cardiac arrest due to VF, and have NYHA Class II or III heart failure, LVEF $\leq 35\%$. Additionally, patients must not have:

- Had a CABG, or PCI with angioplasty and/or stenting, within the past 3 months; or,
- Had an MI within the past 40 days; or,
- Clinical symptoms and findings that would make them a candidate for coronary revascularization.

For these patients identified in B3, a formal shared decision making encounter must occur between the patient and a physician (as defined in Section 1861(r)(1) of the Act) or qualified nonphysician practitioner (meaning a physician assistant, nurse practitioner, or clinical nurse specialist as defined in §1861(aa)(5) of the Act) using an evidence-based decision tool on ICDs prior to initial ICD implantation. The shared decision making encounter may occur at a separate visit.

4. Patients who have severe, non-ischemic, dilated cardiomyopathy but no personal history of sustained VT or cardiac arrest due to VF, and have NYHA Class II or III heart failure, LVEF $\leq 35\%$, been on optimal medical therapy for at least 3 months. Additionally, patients must not have:

- Had a CABG or PCI with angioplasty and/or stenting, within the past 3 months; or,
- Had an MI within the past 40 days; or,
- Clinical symptoms and findings that would make them a candidate for coronary revascularization.

For these patients identified in B4, a formal shared decision making encounter must occur between the patient and a physician (as defined in Section 1861(r)(1) of the Act)

or qualified nonphysician practitioner (meaning a physician assistant, nurse practitioner, or clinical nurse specialist as defined in §1861(aa)(5) of the Act) using an evidence-based decision tool on ICDs prior to initial ICD implantation. The shared decision making encounter may occur at a separate visit.

5. Patients with documented, familial or genetic disorders with a high risk of life-threatening tachyarrhythmias (sustained VT or VF), to include, but not limited to, long QT syndrome or hypertrophic cardiomyopathy.

For these patients identified in B5, a formal shared decision making encounter must occur between the patient and a physician (as defined in Section 1861(r)(1) of the Act) or qualified nonphysician practitioner (meaning a physician assistant, nurse practitioner, or clinical nurse specialist as defined in §1861(aa)(5) of the Act) using an evidence-based decision tool on ICDs prior to initial ICD implantation. The shared decision making encounter may occur at a separate visit.

6. Patients with an existing ICD may receive an ICD replacement if it is required due to the end of battery life, elective replacement indicator (ERI), or device/lead malfunction.

For each of the 6 covered indications above, the following additional criteria must also be met:

1. Patients must be clinically stable (e.g., not in shock, from any etiology);
2. LVEF must be measured by echocardiography, radionuclide (nuclear medicine) imaging, cardiac magnetic resonance imaging (MRI), or catheter angiography;
3. Patients must not have:
 - Significant, irreversible brain damage; or,
 - Any disease, other than cardiac disease (e.g., cancer, renal failure, liver failure) associated with a likelihood of survival less than 1 year; or,
 - Supraventricular tachycardia such as atrial fibrillation with a poorly controlled ventricular rate.

Exceptions to waiting periods for patients that have had a CABG, or PCI with angioplasty and/or stenting, within the past 3 months, or had an MI within the past 40 days:

Cardiac Pacemakers: Patients who meet all CMS coverage requirements for cardiac pacemakers, and who meet the criteria in this national coverage determination for an ICD, may receive the combined devices in one procedure, at the time the pacemaker is clinically indicated;

Replacement of ICDs: Patients with an existing ICD may receive an ICD replacement if it is required due to the end of battery life, ERI, or device/lead malfunction.

C. Nationally Non-Covered Indications

N/A

D. Other

For patients that are candidates for heart transplantation on the United Network for Organ Sharing (UNOS) transplant list awaiting a donor heart, coverage of ICDs, as with cardiac

resynchronization therapy, as a bridge-to-transplant to prolong survival until a donor becomes available, is determined by the local Medicare Administrative Contractors (MACs).

All other indications for ICDs not currently covered in accordance with this decision may be covered under Category B Investigational Device Exemption (IDE) trials (42 CFR 405.201).

(This NCD last reviewed February 2018.)

Claims Processing Instructions

[TN 13483 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

12112

Coverage Transmittal Link

<https://www.cms.gov/files/document/r12112ncd.pdf> 

Revision History

01/2024 - Transmittal 12318 issued October 19, 2023, is being rescinded and replaced by Transmittal 12441, dated January 3, 2024, to add FISS to BR 13390.11 and to add clarifying verbiage to business requirement 13390.12. All other information remains the same. ([TN 12441](#) ) (CR13390)

10/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12318](#) ) (CR13390)

04/2023 - Transmittal 11884 issued March 01, 2023, is being rescinded and replaced by Transmittal 11952, dated, April 12, 2023, to remove the A/B MACs (Part A) and FISS from BR 13070.1 and to revise the NCD 20.4 Implantable Automatic Defibrillators (ICDs) spreadsheet. All other information remains the same. ([TN 11952](#) ) (CR13070)

03/2023 - Transmittal 11832 issued February 02, 2023, is being rescinded and replaced by Transmittal 11884, dated, March 1, 2023 to (1) Business Requirement (BR) 13070.4, NCD 150.3 - delete three non-covered CPT codes added in error and remove reference to ALERT M38; (2) BR 13070.1, NCD 20.4 - clarify that C codes are only payable in the ASC setting; and (3) NCD 220.13 spreadsheet - correct effective date of MSN 21.11 to December 31, 2022 to align with BR 13070.7. All other information remains the same. ([TN 11884](#) ) (CR13070)

02/2023 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11832](#) ) (CR12960)

11/2022 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11676](#)) (CR12960)

04/2022 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11391](#)) (CR12606)

04/2022 - Transmittal 11264, dated February 10, 2022, is being rescinded and replaced by Transmittal 11342, dated, April 6, 2022 to (1) revise BR 12606.10 instructions for NCD 110.24, (2) BR12606.2, fix typo in NCD 160.18 spreadsheet ICD-10 G40.384, which should be G40.834, and, (3) revise implementation verbiage (no changes to the actual implementation date). All other information remains the same. ([TN 11342](#)) (CR12606)

02/2022 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11264](#)) (CR12606)

01/2022 - Transmittal 11068, dated October 21, 2021, is being rescinded and replaced by Transmittal 11179, dated, January 12, 2022 to revise the attachment for NCD 110.24, CAR-T, to add business requirement 12480.10.1 by adding generic unspecified procedure codes, to clarify coverage and claims processing in the policy section and to review the implementation date. All other information remains the same. ([TN 11179](#)) (CR12480)

10/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11068](#)) (CR12480)

09/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11025](#)) (CR12399)

08/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10963](#)) (CR12399)

2/2019 - Transmittal 211, dated December 13, 2018, is being rescinded and replaced by Transmittal 213 dated, February 15, 2019 to change the implementation date from February 26, 2019 to March 26, 2019. All other information remains the same.

12/2018 - Transmittal 209, dated November 21, 2018, is being rescinded and replaced by Transmittal 211, dated, December 13, 2018, to add a clarifying note to the policy section of

the Business Requirements document that was inadvertently left out. All other information remains the same. ([TN 211](#)) (CR10865)

11/2018 - This Change Request (CR) and Publication (Pub.) 100-03 Medicare NCD Manual reflects the Agency's final decision dated February 15, 2018, regarding the reconsideration of NCD 20.4, Implantable Cardiac Defibrillators. A subsequent CR will be released at a later date that contains a Pub.100-04 Claims Processing Manual update with accompanying instructions. Until that time, the Medicare Administrative Contractors (MACs) shall be responsible for implementing NCD 20.4. ([TN 209](#)) (CR10865)

06/2016 - Transmittal 1665, dated May 13, 2016, is being rescinded and replaced by Transmittal 1672 to: (1) 9631.1: Remove FISS responsibility and 1st sentence; (2) 9631.2: Remove additional procedure codes, including 0075T, 0076T; (3) 9631.4: Revise descriptor of dx L59.8; (4) 9631.6: Add deletion of dx C49.10, C65.9, remove deletion of dx C54.9, remove deletion of invalid dx C47.90; (5) 9631.8: Remove deletion of invalid dx C51.29, replace with deletion of dx C50.029; (6) 9631.9: Add deletion of 0V504ZZ, 0V500ZZ, override capability, and contractor discretion verbiage. All other information remains the same. ([TN 1672](#)) (CR9631)

05/2016 - This change request (CR) is the 7th maintenance update of ICD-10 conversions and other coding updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, CR9087, CR9252, and CR9540. Some are the result of revisions required to other NCD-related CRs released separately. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent, quarterly releases as needed. No policy-related changes are included with these updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1665](#)) (CR9631)

03/2013 - CMS translated the information for this policy from ICD-9-CM/PCS to ICD-10-CM/PCS according to HIPAA standard medical data code set requirements and updated any necessary and related coding infrastructure. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 10/07/2013 Effective date: 10/1/2015. ([TN 1199](#)) (CR 8197)

3/2005 - Expanded national coverage by including new indications. Effective and implementation dates 01/27/2005. ([TN 29](#)) (CR 3604)

08/2003 - Expanded coverage for coronary artery disease with a documented prior myocardial infarction (MI), a measured left ventricular ejection fraction ≤ 0.35 , and inducible, sustained ventricular tachyarrhythmia or ventricular fibrillation at electrophysiology study; or documented prior MI and a measured left ventricular ejection fraction ≤ 0.30 and a QRS duration of >120 milliseconds. All other indications for defibrillators not otherwise specified remain noncovered except when furnished in accordance with FDA-approved protocols governing Category B Investigational Device Exemption; clinical trials as stated in 42 FR Part 405. Effective and implementation dates 10/01/2003. ([TN 173](#)) (CR 2880)

04/1999 - Expanded coverage for additional types of cardiac patients. Effective and implementation dates 07/01/1999. (TN 115) (CR 825)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Implantable Cardioverter Defibrillators \(ICDs\) \(CAG-00157N\)](#)
- [First reconsideration for Implantable Defibrillators - Clinical Trials \(CAG-00157R\)](#)
- [Second reconsideration for Implantable Defibrillators \(CAG-00157R2\)](#)
- [Third reconsideration for Implantable Defibrillators \(CAG-00157R3\)](#)
- [Fourth reconsideration for Implantable Cardioverter Defibrillators \(CAG-00157R4\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Implantable Cardioverter Defibrillators (ICDs)	5	07/31/2023 - N/A	You are here
Implantable Cardioverter Defibrillators (ICDs)	4	02/15/2018 - 07/31/2023	View
Implantable Automatic Defibrillators	3	01/27/2005 - 02/15/2018	View
Implantable Automatic Defibrillators	2	10/01/2003 - 01/27/2005	View
Implantation of Automatic Defibrillators	1	07/01/1999 - 10/01/2003	View